

A Cosmological Model Explaining Matter and Dark Matter Formation, Galactic Rotation Curve Anomalies, the Initial Low-Entropy State of the Universe, and the Mechanism of Accelerated Expansion

Abstract : This paper attempts to present a cosmological evolution model that is consistent with both general relativity and quantum field theory. It aims to explain various cosmological issues, including the formation processes of matter and dark matter, the anomaly in galaxy rotation curves, the specific physical picture of the universe's initial low-entropy state, and why the universe is currently in an accelerated expansion phase. This article is a part of the preprint "Multiplicative Entropy Encodes Time-Causality; Gauge Mediates Mass-Gravity; Frequency Embodies Metric Evolution. Section 9.1 Zenodo. <https://doi.org/10.5281/zenodo.15278220>".

Keywords:

Cosmology Model;dark matter formation; matter formation ;anomaly in galaxy rotation curves;universe's initial low-entropy state; accelerated expansion mechanism;

Introduction: This paper presents a novel cosmological model based on the concept of quantized elastic spacetime (SEQ) , aiming to unify the principles of general relativity and quantum field theory within a single theoretical framework. The model interprets spacetime not merely as a geometric entity, but as a quantum-elastic medium composed of sub-Planck-scale components, capable of storing and transferring energy through compression, stretching, and spinor interactions. The motivation for this model arises from several unresolved issues in modern cosmology, including the initial low-entropy state of the universe, the anomalies in galaxy rotation curves, the nature of dark matter, and the mechanism behind the current accelerated expansion of the universe. Traditional models often rely on the introduction of dark matter and dark energy to explain these phenomena. However, these components remain elusive and lack direct observational evidence.

In contrast, the quantized elastic spacetime model offers a physically intuitive and geometrically grounded explanation for these cosmological puzzles. By analyzing the energy transformations among resonant kinetic energy, spin kinetic energy, and elastic potential energy across different cosmic evolution stages, the model provides a coherent description of mass generation, matter-energy conversion, gravitational effects, and the evolution of entropy.

Furthermore, the model introduces a Higgs-like mechanism—termed the space network spinor–SEQ fixed chirality coupling—which explains how spatial compression states are locked into mass states during the early inflationary phase. This mechanism also provides a physical basis for the formation of dark matter and the observed gravitational effects in galaxies, without requiring exotic new particles.

By integrating thermodynamics, quantum field theory, and general relativity, this work offers a self-consistent and testable cosmological framework, shedding new light on the fundamental nature of spacetime, matter, and cosmic evolution.

1 Postulates: Consider the universe as a quantized elastic spacetime model,This is consistent with both general relativity and quantum field theory.

2 Initial low-entropy state of the universe.

Before the Big Bang, the universe was in a highly compressed state. The elastic potential energy was divided into two components: one was spatial compression potential energy, and the other was spatial twistor potential energy(Space network spinor). The spatial twistor potential energy was concentrated in an extremely small region of space — this constituted the initial low-entropy state. After the Big Bang, the twistor potential energy from this tiny region was released through conversion into SEQ resonant kinetic energy and SEQ spin kinetic energy, and this energy was then conducted to other regions of space. This process represents the specific mechanism of entropy increase.

3 Universe evolution phases and Why is the universe currently in an accelerated expansion phase?

Stage-Phase	Stage Name	Process	Universe State	Thermodynamic Characteristics
0-Compression	Pre-Big Bang Initial State	The universe's SEQ network is highly compressed, with resonant frequencies close to zero. The initial low-entropy state may be reflected in a few SEQ having particularly high energy, while most have low energy.	High-energy Aggregation State	Low entropy
1-Compression	Compression Potential Energy → Kinetic Energy	Elastic compression potential energy is released and converted into cosmic expansion kinetic energy	Accelerating Expansion	Low entropy, high energy concentration, rapid entropy increase
2-Stretching	Kinetic Energy → Tension Potential Energy	Expansion kinetic energy is converted into tension potential energy	Decelerating Expansion	Increasing entropy
3-Stretching	Tension Potential Energy → Kinetic Energy	Tension potential energy is released and converted into contraction kinetic energy	Accelerating Contraction	Entropy continues to increase
4-compression	Kinetic Energy → Compression Potential Energy	Contraction kinetic energy is converted into compression potential energy	Decelerating Contraction	Entropy continues to increase
5-repeated Oscillation → Equilibrium Oscillation	Energy Homogenization → Equilibrium Oscillation	In each cycle, the energy distribution becomes more uniform, with no obvious concentrated	Approaching Equilibrium State	Entropy approaches maximum, oscillating universe in thermal

Stage-Phase	Stage Name	Process	Universe State	Thermodynamic Characteristics
		states remaining		equilibrium
This process does not collapse back to the initial birth configuration of universe, nor does it reduce entropy—since the entropy increasing trend remains invariant under expansion or contraction, the homogenization of energy distribution is an irreversible process, until entropy reaches its maximum value.				

1	Potential energy is stored in elastic bonds composed of sub-Planck scale components.
2	In this model the energy of SEQ m_i equals the SEQ resonant kinetic energy plus the SEQ spin kinetic energy plus the elastic potential energy assigned to this SEQ from its adjacent elastic bonds manifested as frequency suppression.
3	Chapter 10 will discuss that mass formation is mainly due to spin locking the spatial compression state and the key factor is the coupling confinement potential between the network spinor and the SEQ fixed chirality spin.

- E global SEQ network =Kresonant +K spin +Uelastic;
- Uelastic=U compress-stretch +U twistor; (Space network spinor)
- Utwistor (Space network spinor) converts into Kresonant +K spin; embodied as space network spinor
- Ucompress-stretch converts into Kresonant

4 The mass formation and matter-energy conversion mechanisms in different stages of cosmic evolution:

- First stage of cosmic expansion: mass generation dominates.

According to this quantized elastic space-time model, mass is formed when compressed states are locked by the coupling of SEQ network spinors and the fixed chirality spin of SEQ. The formation of mass states depends on the degree of spatial compression and the intensity of SEQ network spinor. Since the universe as a whole is in a highly compressed phase during this stage, and SEQ network spinor is concentrated on a local zone, it is consistent that mass is formed. In the first stage, the entire cosmic space is generally in a highly compressed state, and during the initial inflationary period, the compression is even more pronounced. Due to the space network spinor--SEQ fixed chirality coupling locking mechanism (similar to the Higgs mechanism, introduced in Chapter 10), the highly concentrated spinor of local zone causes certain compressed spatial states to be locked into mass states. Therefore, mass generation dominates in this stage,

especially during the early inflationary phase, when the spatial compression level is higher and better matches the compression required for mass state formation.

- Second and third stages of cosmic evolution:

transformation from mass states to energy states dominates, with possible new mass state formation during the process.

In these stages, the universe as a whole enters a stretched phase. According to this model, mass states become unstable due to the stretching of the space itself. The conditions for mass state become weaker, leading to more mass states breaking down. During this phase, new elements—mass states—may be generated within massive celestial bodies.

- Fourth stage of cosmic evolution : the universe re-enters a compressed phase.

However, at this stage, the distribution of SEQ network spinor distribution has become more uniform, and the effect of the Higgs-like mechanism is weaker. Although new mass states may still form, the overall rate of mass generation is significantly lower than in the first stage.

- Fifth stage of cosmic evolution:

repeated oscillations leading into thermal equilibrium oscillation.

As the network energy becomes increasingly uniformly distributed—an irreversible entropy increase process—the proportion of mass states in the universe gradually decreases through repeated oscillations. Eventually, all mass states are transformed into energy states, meaning the spin kinetic energy distribution of the SEQ network becomes uniform, and the universe enters oscillation in a thermal equilibrium state.

5. Anomalies in galaxy rotation curves.

This mechanism, in which high spatial compression collaborates with space network spinor to generate mass, may also explain the observed **anomalies in galaxy rotation curves**. Specifically, the early universe featured a high concentration of network spinor. This intense network spinor concentration, coupled with the extreme spatial compression, facilitated the formation of mass. Consequently, the resulting mass structures and their surrounding space likely retain higher-than-expected spinor, which could account for the observed rotational dynamics in galaxies.

6 Dark matter formation.

Dark matter formation is also dominated by a synergistic locking mechanism involving spatial compression states and a Higgs-like mechanism (space network spinor and SEQ fixed-chirality spin coupling), similar to the mass generation mechanism in the early stages of cosmic expansion. However, dark matter formed at a later stage than ordinary mass. The spatial compression level during this phase was insufficient to reach the threshold required for mass generation. Moreover,

by this time, the increase in entropy had led to a more uniform distribution of space network spinor, reducing the strength of the spinor field in the Higgs-like mechanism to a level insufficient to lock high-compression states. This is the fundamental difference between dark matter and mass states. In essence, dark matter is also a form of locked spatial compression state, but with a much lower compression strength compared to mass states. Nevertheless, it still exerts a stretching effect on the surrounding space, manifesting as a gravitational field.

7 Gravitation field mechanism.

Why is there a global gravitational tension-restoring potential in the first stage of cosmic expansion, even though the entire space is in a compressed phase? Because the compression level within mass structures is significantly higher than that of the surrounding space, leading to a large gradient in spatial deformation, which gives rise to a relative tension-restoring potential — the gravitational potential.

This explanation is self-consistent and visually clear, especially in providing a concrete physical image and mechanism for the large-scale generation of mass states in the early inflationary universe.

For more details, please refer to the original text:

<https://doi.org/10.5281/zenodo.15278220>